

Networking Services

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Networking Basics

- Networking is the interconnected system of devices, often computers, that can communicate and share data and resources.
- It's a process that enables devices to exchange information and collaborate, whether it's sharing a printer, accessing the internet, or collaborating on a document.
- Think of it as a network of interconnected nodes, where each node can be a computer, phone, printer, or any device that can send and receive data.

IP Addresses:

- Every device on a network (e.g., a computer) has a unique IP address, like a postal address, which allows other devices to find and communicate with it.

Protocols (TCP/IP):

- Protocols are a set of rules that define how data is transmitted and received on a network. TCP/IP is a standard protocol suite used extensively on the internet.
- TCP (Transmission Control Protocol) ensures reliable delivery of data, including retransmission of lost packets.
- IP (Internet Protocol) handles the addressing and routing of data packets across the network.

Packets

- When data is transmitted across a network, it is often divided into smaller units called packets.
- These packets contain the data, the source and destination IP addresses, and other information needed for routing.
- The network then delivers these packets from one device to another, and they are reassembled at the destination to create the original data.

Why is Networking Important

Resource Sharing:

- Networking allows devices to share resources like printers, storage, and applications.

Communication:

- It enables communication between devices, such as sending emails, browsing the internet, and collaborating on documents.

Internet Access:

- The internet itself is a large network that connects billions of devices worldwide, enabling communication and access to vast amounts of information.

Data Exchange:

- Networking is essential for transferring data between devices, whether it's uploading files, streaming video, or downloading software.

Types of Networks

LAN (Local Area Network): A network within a limited area, like a home or office.

MAN (Metropolitan Area Network): A network covering a larger geographical area, like a city.

WAN (Wide Area Network): A network covering a very large geographical area, like the internet.



OSI Model Overview



Application Layer: Human-computer interaction layer, where applications can access the network services

Presentation Layer : Ensures that data is in a usable format and is where data encryption occurs.

Session Layer : Maintains connections and is responsible for controlling ports and sessions

Transport Layer: Transmits data using transmission protocols including TCP and UDP.

Network Layer: Decides which physical path the data will take.

Data Link Layer: Defines the format of data on the network.

Physical Layer: Transmits raw bit stream over the physical medium

OSI Layer 3 (Network)

The Network Layer provides the functional and procedural means of transferring variable length data sequences from a source to a destination via one or more networks, while maintaining the quality of service requested by the Transport Layer.

The Network Layer performs

- network routing functions,
- perform fragmentation and reassembly,
- report delivery errors.

Routers operate at this layer—sending data throughout the extended network and making the Internet possible.

OSI Layer 3 (Transport)

The Transport Layer provides transparent transfer of data between end users, providing reliable data transfer services to the upper layers.

The Transport Layer controls the reliability of a given link through flow control, segmentation/desegmentation, and error control.

Feature Name	TP0	TP1	TP2	TP3	TP4
Connection oriented network	Yes	Yes	Yes	Yes	Yes
Connectionless network	No	No	No	No	Yes
Concatenation and separation	No	Yes	Yes	Yes	Yes
Segmentation and reassembly	Yes	Yes	Yes	Yes	Yes
Error Recovery	No	Yes	No	Yes	Yes
Reinitiate connection (if an excessive number of PDUs are unacknowledged)	No	Yes	No	Yes	No
multiplexing and demultiplexing over a single virtual circuit	No	No	Yes	Yes	Yes
Explicit flow control	No	No	Yes	Yes	Yes
Retransmission on timeout	No	No	No	No	Yes
Reliable Transport Service	No	Yes	No	Yes	Yes

IP Addressing & CIDR

➤ What is an IP Address?

An IP address is a unique global address for a network interface

- Exceptions:
- Dynamically assigned IP addresses (☐ DHCP)
- IP addresses in private networks (☐ NAT)

An IP address:

- is a 32 bit long identifier
- encodes a network number (network prefix) and a host number

IP Addressing & CIDR Conti

- The network prefix identifies a network and the host number identifies a specific host (actually, interface on the network).

`network prefix`

`host number`

- **How do we know how long the network prefix is?**
 - Before 1993: The network prefix is implicitly defined (see class-based addressing) or
 - After 1993: The network prefix is indicated by a netmask.

Classful IP Addresses (Until 1993)

When Internet addresses were standardized (early 1980s), the Internet address space was divided up into classes:

- Class A: Network prefix is 8 bits long
- Class B: Network prefix is 16 bits long
- Class C: Network prefix is 24 bits long

Each IP address contained a key which identifies the class:

- Class A: IP address starts with “0”
- Class B: IP address starts with “10”
- Class C: IP address starts with “110”

Problem with Classful IP Addresses

- By the early 1990s, the original classful address scheme had a number of problems

Flat address space. Routing tables on the backbone Internet need to have an entry for each network address. When Class C networks were widely used, this created a problem. By the 1993, the size of the routing tables started to outgrow the capacity of routers.

Other problems:

- Too few network addresses for large networks

Class A and Class B addresses were gone

- Limited flexibility for network addresses:

Class A and B addresses are overkill (>64,000 addresses)

Class C address is insufficient (requires 40 Class C addresses)

CIDR- Classless Interdomain Routing

- IP backbone routers have one routing table entry for each network address:
 - ❑ With subnetting, a backbone router only needs to know one entry for each Class A, B, or C networks
 - ❑ This is acceptable for Class A and Class B networks
 - 27 = 128 Class A networks
 - 214 = 16,384 Class B networks
 - ❑ But this is not acceptable for Class C networks
 - 221 = 2,097,152 Class C networks

CIDR- Classless Interdomain Routing

- In 1993, the size of the routing tables started to outgrow the capacity of routers.
- Consequence: The Class-based assignment of IP addresses had to be abandoned .

Goals:

- New interpretation of the IP address space
- Restructure IP address assignments to increase efficiency
- Permits route aggregation to minimize route table entries

CIDR (Classless Interdomain routing)

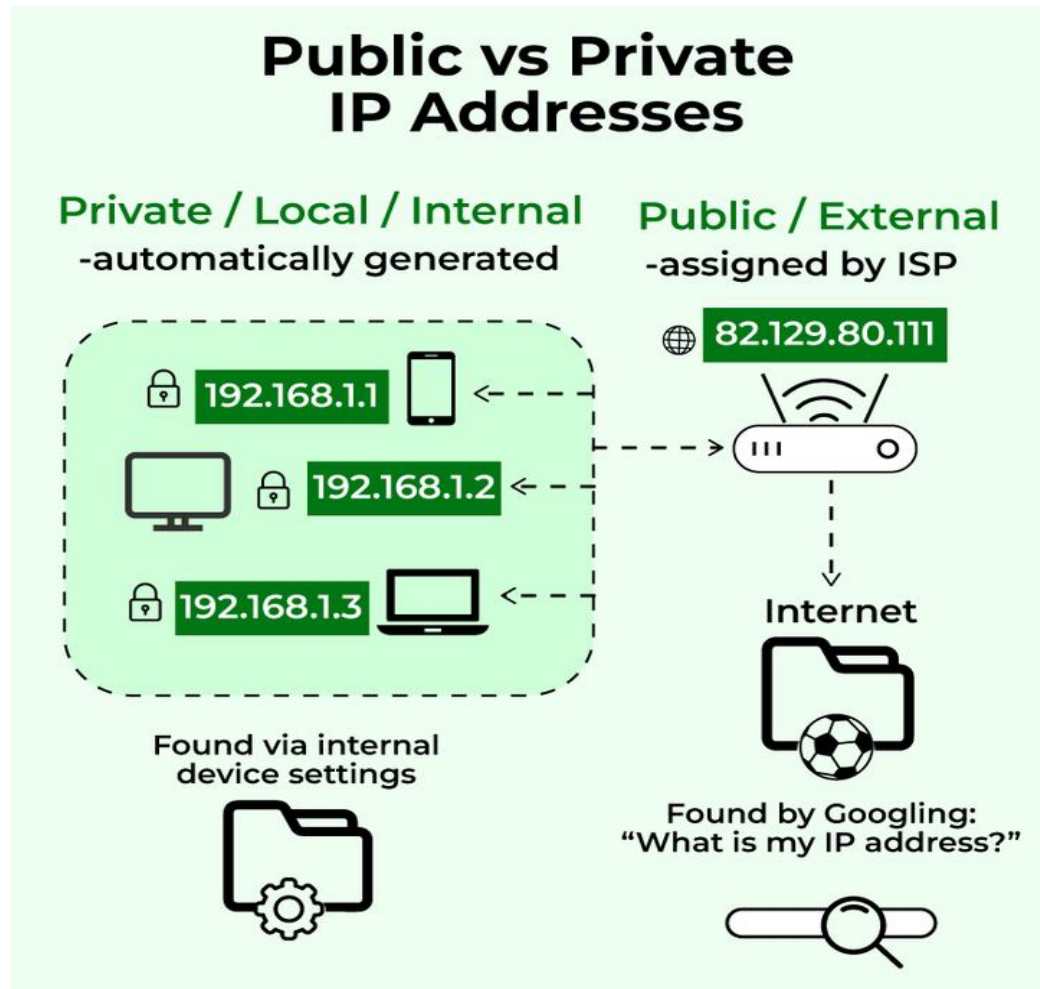
- abandons the notion of classes
- Key Concept: The length of the network prefix in the IP addresses is kept arbitrary
- Consequence: Size of the network prefix must be provided with an IP address
- CIDR notation of an IP address:

192.0.2.0/18

"18" is the prefix length. It states that the first 18 bits are the network prefix of the address (and 14 bits are available for specific host addresses)

- CIDR notation can replace the use of subnetmasks (but is more general)
 - IP address 128.143.137.144 and subnetmask 255.255.255.0 becomes
128.143.137.144/24
- CIDR notation allows to drop trailing zeros of network addresses:
- 192.0.2.0/18 can be written as 192.0.2/18

Private Vs Public IP Addresses



Private Vs Public IP Addresses

Private IP Address

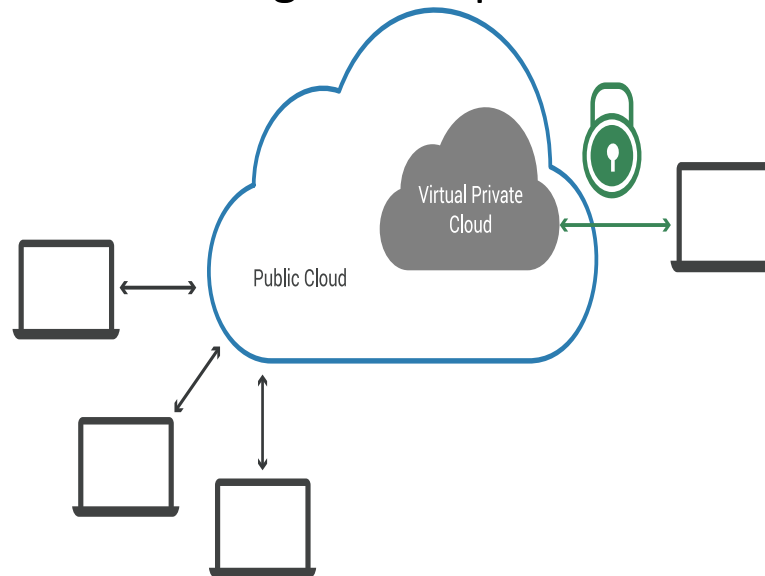
- The scope of Private IP is local.
- It is used to communicate within the network.
- Private IP addresses of the systems connected in a network differ in a uniform manner.
- It works only on LAN.
- It is used to load the network operating system.
- It is available free of cost.
- Private IP can be known by entering “ipconfig” on the command prompt.

Public IP Address

- The scope of Public IP is global.
- It is used to communicate outside the network.
- Public IP may differ in a uniform or non-uniform manner.
- It is used to get internet service.
- It is controlled by ISP.
- It is not free of cost.
- Public IP can be known by searching “what is my ip” on Google.
- Public IP does not require a network translation

Virtual Private Cloud(VPC)

- A virtual private cloud (VPC) is a secure, isolated private cloud hosted within a public cloud. VPC customers can run code, store data, host websites, and do anything else they could do in an ordinary private cloud, but the private cloud is hosted remotely by a public cloud provider.
- It is a service that enables the users to gain complete control over the virtual environment.



Virtual Private Cloud(VPC)

- It allows the users to select their own IP address, create subnets, configure route tables, and select the network gateways as well.
- Imagine a public cloud as a crowded restaurant, and a virtual private cloud as a reserved table in that crowded restaurant. Even though the restaurant is full of people, a table with a "Reserved" sign on it can only be accessed by the party who made the reservation. Similarly, a public cloud is crowded with various cloud customers accessing computing resources – but a VPC reserves some of those resources for use by only one customer.

Public Vs Private Cloud

- A public cloud is shared cloud infrastructure. Multiple customers of the cloud vendor access that same infrastructure, although their data is not shared – just like every person in a restaurant orders from the same kitchen, but they get different dishes. Public cloud service providers include AWS, Google Cloud Platform, and Microsoft Azure, among others.
- A private cloud, however, is single-tenant. A private cloud is a cloud service that is exclusively offered to one organization. A virtual private cloud (VPC) is a private cloud within a public cloud; no one else shares the VPC with the VPC customer.

How is VPC Isolated in Public Cloud?

- A VPC isolates computing resources from the other computing resources available in the public cloud. The key technologies for isolating a VPC from the rest of the public cloud are:
- **Subnets:** A subnet is a range of IP addresses within a network that are reserved so that they're not available to everyone within the network, essentially dividing part of the network for private use. In a VPC these are private IP addresses that are not accessible via the public Internet, unlike typical IP addresses, which are publicly visible.

How is VPC Isolated in Public Cloud?

- **VLAN:** A LAN is a local area network, or a group of computing devices that are all connected to each other without the use of the Internet. A VLAN is a virtual LAN. Like a subnet, a VLAN is a way of partitioning a network, but the partitioning takes place at a different layer within the OSI model (layer 2 instead of layer 3).
- **VPN:** A virtual private network (VPN) uses encryption to create a private network over the top of a public network. VPN traffic passes through publicly shared Internet infrastructure – routers, switches, etc. – but the traffic is scrambled and not visible to anyone.

How is VPC Isolated in Public Cloud?

- A VPC will have a dedicated subnet and VLAN that are only accessible by the VPC customer. This prevents anyone else within the public cloud from accessing computing resources within the VPC – effectively placing the "Reserved" sign on the table. The VPC customer connects via VPN to their VPC, so that data passing into and out of the VPC is not visible to other public cloud users.
- Some VPC providers offer additional customization with:
- **Network Address Translation (NAT):** This feature matches private IP addresses to a public IP address for connections with the public Internet. With NAT, a public-facing website or application could run in a VPC.
- **BGP route configuration:** Some providers allow customers to customize BGP routing tables for connecting their VPC with their other infrastructure

Default vs. Custom VPC

Default VPC

- A Default VPC is a pre-configured, automatically created virtual network in Amazon Web Services (AWS) for each region.
- Default VPCs offer a quick and easy way to get started
- Default VPCs use a predefined 172.31.0.0/16 CIDR block
- Default VPCs typically include internet access via an internet gateway.
- Default VPCs are suitable for basic deployments, testing, and quick projects

Custom VPC

- A Custom VPC is one that you create and configure yourself.
- Custom VPCs provide greater control and flexibility for advanced networking needs.
- Custom VPCs allow you to choose your own CIDR block and subnets.
- Custom VPCs require you to configure internet access if needed.
- Custom VPCs are better for production environments, complex network setups, and when you need specific configurations.

Default vs. Custom VPC

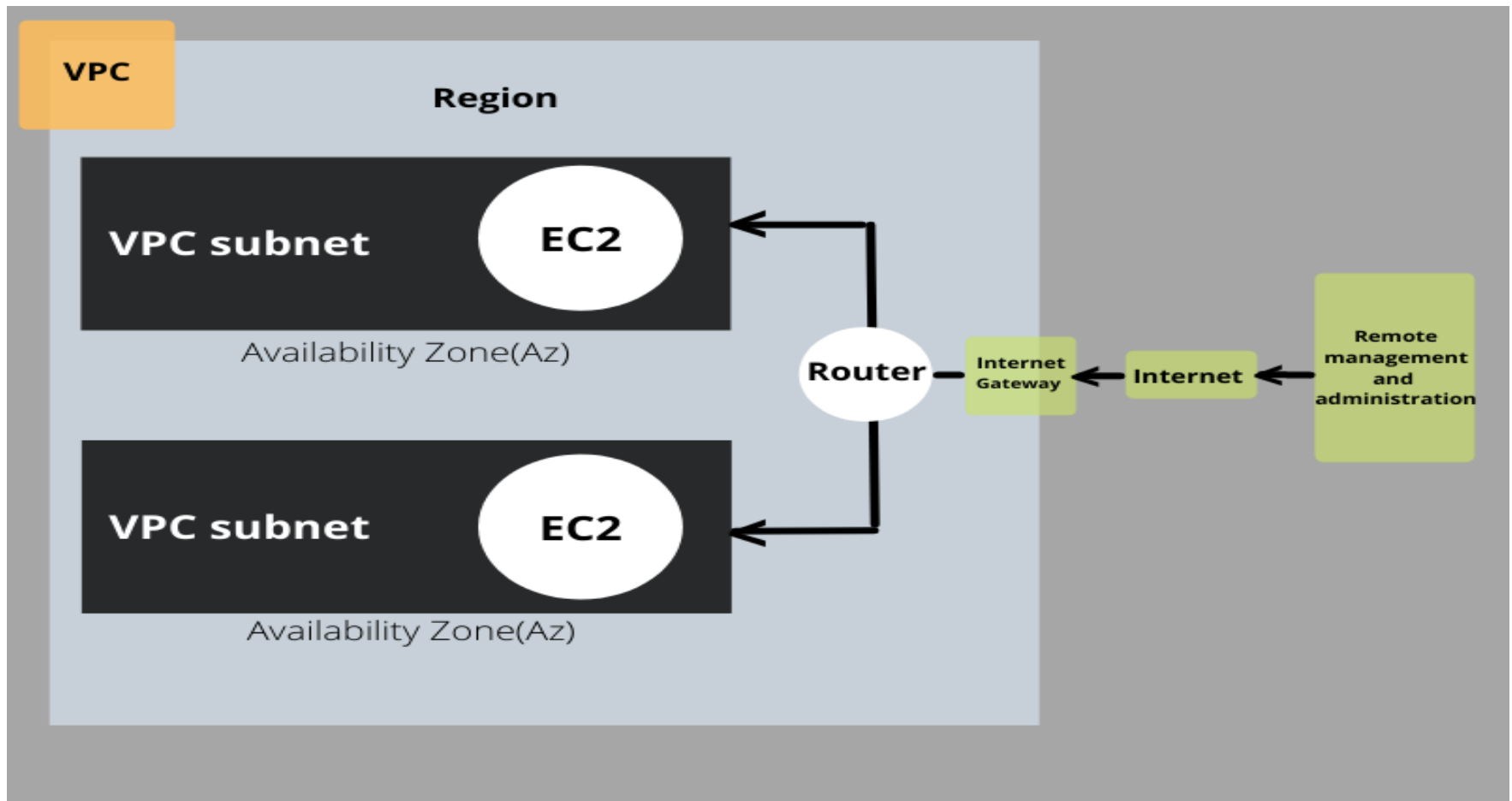
Default VPC

- You cannot peer two Default VPCs.
- Suitable for basic deployments, testing, and quick projects where you don't need custom IP addresses or specific network configurations.

Custom VPC

- Custom VPCs can be peered with each other, other VPCs, or on-premises networks.
- Preferred for production environments, complex networking needs, advanced security configurations, and when you need to control IP address ranges, subnets, and routes.

VPC Components



VPC Components

- **Subnet:** A subnet in VPC is something a range of IP addresses. It is a section of a VPC that can contain resources such as Amazon EC2 services and shares a common address component. Public Subnet where resources are exposed to the internet through Internet Gateway and Private Subnet where resources are not exposed to the outside world.
- **Route Table:** They are the set of rules used to decide where the network traffic has to be managed. It specifies the destination i.e IP address and target. The target can be Internet gateway, NAT gateway, Virtual private gateway, etc. With the use of route tables, users can determine where the network traffic will be directed from your subnet or gateway.

VPC Components

- **Virtual Private Gateway:** It is the VPN(Virtual Private Network) hub on the Amazon side of the VPN connection to have a secure transaction. Users can attach it to the VPC from which they want to create the VPN connection.
- **NAT Gateway:** Network Address Translation (NAT) Gateway is used when higher bandwidth, availability with lesser management effort is required. It updates the routing table of the private subnet such that it sends the traffic to the NAT gateway. It supports only UDP, TCP, and ICMP protocols.

VPC Components

- **VPC Peering:** A VPC peering connection allows you to route traffic between two Virtual Private Clouds using IPv4 or IPv6 private addresses. Users can create a VPC peering connection between their own VPC with a VPC in another AWS account. This connection helps you to smoothly transfer the data.
- **Security Groups:** It consists set of firewalls rules that control the traffic for your sample. You can have a single security group associated with multiple instances.

VPC Components

- **Elastic IP:** It is a static IP address which is a reserved public IP address that can be assigned to any Instance in a particular region and never changes.
- **Network Access Control Lists (NACL):** It is an optional layer of security for your VPC that acts as a firewall for controlling traffic in and out of one or more subnets. It adds an additional layer of security to your VPC.
- **Customer Gateway:** VPN connection links your network (or data) to your Amazon VPC (virtual private cloud). A customer gateway is a presenter on your side of that connection. It can be a physical or software appliance.

VPC Components

- **Network Interface:** It's a connection between private and public networks. Network traffic is automatically shifted to the new instance if you move it from one instance to the other.
- **VPC Endpoints:** It allows VPC to make a connection with other services of AWS without using the internet. They are of two types, Interface endpoints, and Gateway endpoints. They are scaled, redundant, and highly available VPC components.
- **IP addressing:** With the IP Addressing, you can assign your VPCs and subnets, the IPv4 addresses and IPv6 addresses.

VPC Peering and Transit Gateway

- When you allow to connect one VPC with another VPC through some of direct network route and using some private IP addresses, then this connection is called VPC Peering. and Instances will behave as if they are on the same private network. Users can peer between the regions. From VPC Peering, users can easily peer the VPC's with another AWS accounts and also users can peer the VPCs in the same account.
- It enables direct communication between the peered VPCs as if they are part of the same network. Suppose you have multiple AWS accounts. In this case, you can connect the VPCs of the accounts and create a file-sharing network

VPC Peering and Transit Gateway

- AWS Transit Gateway is an AWS service that helps us to establish the connectivity between Amazon Virtual Private Clouds (VPCs), on-premises networks, and remote networks.
- It acts as a central hub that allows communication between connected networks. It works as a virtual router to connect the AWS Virtual Private Clouds and VPN connections.
- Transit Gateway is scalable, which means that it can accommodate a large number of requests easily. In other words, it simplifies the networking architecture and enables efficient traffic routing between different environments.

Which to choose VPC Peering OR Transit Gateway

- VPC Peering is useful for setting up a simple network scenario with a limited number of VPCs. In such cases, the direct connections between pairs of VPCs are sufficient to route the traffic.
- Also, you can use VPC Peering if your primary requirement is resource sharing and collaboration between specific VPCs without the need for a centralized hub.
- You need to use VPC Peering if you face challenges in the complex hub-and-spoke model. Thus, it is useful for the cases in which we just need a straightforward and direct connectivity between two VPCs.

Which to choose VPC Peering OR Transit Gateway

- Transit Gateway is suitable if you need a scalable network architecture to connect multiple VPCs, on-premises networks, and remote networks in a hub-and-spoke model.
- You should choose the Transit Gateway if You need cross-region connectivity. Thus, you can connect VPCs in different regions with Transit Gateway.
- You should select Transit Gateway if your cloud infrastructure requires transitive routing. So, you can use it to enable the inter-VPC communication even when they are not directly peered.

Subnet in VPC

- A subnet (or subnetwork) is a smaller, logical division of a larger network.
- It's essentially a network within a network, allowing for more efficient use of IP addresses and better network organization.
- Subnets are created through a process called subnetting, which involves dividing a larger IP network address space into multiple, smaller address spaces
- Subnets are used to logically segment a VPC into smaller, more manageable network sections. This helps in isolating resources and controlling access to them.

Subnet in VPC

- In a Virtual Private Cloud (VPC), a subnet is a range of IP addresses within the VPC. Subnets can be either public or private, each with distinct characteristics.
- Public subnets have a direct route to an Internet gateway, enabling resources within them to directly access the internet. Private subnets, however, do not have this direct route and require a Network Address Translation (NAT) gateway in a public subnet to access the internet.

Public Vs Private Subnet

- Public Subnet:
 - A public subnet has a route table associated with it that includes a route to an Internet Gateway. This allows resources within the subnet to send and receive traffic directly to and from the public internet.
- Private Subnet:
 - A private subnet does not have a direct route to an Internet Gateway. Resources in a private subnet need to use a NAT gateway in a public subnet to access the internet.

Subnet Strategies

Security:

- Using private subnets for sensitive resources, like databases, can significantly enhance security by limiting direct internet access.

Cost:

- Using a NAT gateway for internet access in private subnets can be more cost-effective than having multiple Internet Gateways.

Scalability:

- Using public and private subnets allows for a more scalable and flexible network architecture, enabling resources to be deployed in different zones with varying levels of internet access.

Subnet Strategies

Resource Isolation:

- Separating resources into different subnets helps to isolate traffic and improve resource utilization.
- Multi-AZ deployments are crucial for achieving high availability by distributing resources across multiple Availability Zones (AZs). This ensures that if one AZ experiences a failure, your application can continue running without interruption in another AZ.
- CIDR (Classless Inter-Domain Routing) allocation, while not directly related to Multi-AZ deployments, is best practice for network planning and security.

Route Tables

- In a Virtual Private Cloud (VPC), route tables determine how network traffic is directed between subnets and the internet
 - The main route table is automatically created for the VPC, while custom route tables can be created and associated with specific subnets
 - A default route in the main route table ensures communication within the VPC, while custom routes can be added for traffic to external destinations like the internet or other VPCs

Route Tables : Main vs. Custom

➤ Main Route Table:

Automatically associated with the VPC and includes a default route that allows traffic to be routed within the VPC. It's the default for all subnets not explicitly associated with a custom route table.

➤ Custom Route Tables:

You create these and associate them with specific subnets. This allows for more granular control over traffic routing, such as sending traffic from a private subnet to the internet through a NAT gateway or to other VPCs through VPC peering.

Security Groups and NACL's

Security Groups (SGs) act as a stateful firewall at the instance level, allowing traffic based on predefined rules, while Network ACLs (NACLs) are a stateless firewall at the subnet level, allowing or denying traffic based on rules

Security Groups (SGs):

➤ **Stateful:**

SGs maintain a "state" of the traffic flow. If an inbound rule allows traffic, the corresponding outbound traffic is also automatically allowed, according to Cloudviz.

➤ **Instance Level:**

SGs are associated with individual instances within a VPC.

Security Groups and NACL's

➤ **Allow Rules:**

SGs only allow traffic based on defined rules; they do not have deny rules.

➤ **Multiple SGs:**

An instance can have multiple security groups associated with it, according to Medium.

Network ACLs (NACLs):

- **Stateless:** NACLs operate on a stateless basis, meaning they don't maintain a record of previous traffic.

Security Groups and NACL's

- **Subnet Level:** NACLs are applied at the subnet level within a VPC.
- **Allow/Deny Rules:** NACLs allow you to define both allow and deny rules.
- **Single NACL:** Each subnet can only have one NACL associated with it, according to Medium.
- **Rule Evaluation:** NACL rules are evaluated in order, starting with the lowest number. The first matching rule determines whether the traffic is allowed or denied, according to Medium.

Security Groups Vs NACL's

Security Groups

- Level of application is Instance Level
- Stateful (Allows based on state)
- Allow Rules only
- Multiple security groups can be associated with an instance
- All rules evaluated before deciding on traffic

NACL's

- Level of application is Subnet Level
- Stateless (Separate rules for inbound/outbound)
- Allow and Deny Rules
- One NACL per subnet
- Rules evaluated in order, starting with the lowest number

VPC Security

VPC (Virtual Private Cloud) security involves a range of measures to protect your cloud infrastructure and data. Key aspects include using security groups to control traffic flow, employing Network ACLs (Access Control Lists) to filter traffic at the subnet level, and utilizing VPC Flow Logs for monitoring and auditing traffic.

Security Groups:

These act as virtual firewalls, controlling inbound and outbound traffic to and from individual instances or resources within the VPC. You can define rules to allow or deny traffic based on source IP, port, and protocol.

VPC Security

Network ACLs:

ACLs provide another layer of security at the subnet level, allowing you to filter traffic at the subnet boundary. They complement security groups by offering more granular control over traffic flow to specific subnets.

VPC Flow Logs:

These logs capture information about IP traffic within your VPC, enabling you to monitor and audit network activity. They can help you identify potential security issues and troubleshoot network configurations.

VPC Security

Traffic Mirroring:

This feature allows you to copy network traffic from an elastic network interface and send it to a security appliance or monitoring tool. This can be valuable for intrusion detection and incident response.

AWS Network Firewall:

This service provides managed firewall capabilities, allowing you to filter traffic based on various criteria and protect your VPC from unauthorized access.

VPC Security – Best Practices

Limit Public Access:

Restrict public access to your VPC resources by using security groups and network ACLs to control traffic.

Use Multi-Factor Authentication (MFA):

Enable MFA for all user accounts accessing your VPC to prevent unauthorized access.

Monitor and Audit:

Regularly monitor your VPC traffic using VPC Flow Logs and other monitoring tools to identify potential security breaches.

Update Security Groups and ACLs:

Regularly review and update your security group and network ACL rules to reflect changes in your environment and security needs.

AWS Shield and WAF

AWS Shield and AWS WAF offer distinct protection strategies: AWS Shield defends against DDoS attacks, while AWS WAF protects web applications from various web-based attacks.

Shield focuses on network-level and transport-level attacks, while WAF focuses on application-level attacks. WAF allows for granular control over web requests, enabling the creation of custom rules to block or allow traffic based on various criteria.

AWS Shield

DDoS Protection:

Primarily designed to protect against Distributed Denial of Service (DDoS) attacks, which aim to overload systems with traffic.

Network and Transport Layers:

Offers protection at Layer 3 (network) and Layer 4 (transport) of the OSI model.

Two Levels of Protection:

AWS Shield Standard (included with other AWS services) and AWS Shield Advanced (enhanced protection with additional features).

Global Availability:

AWS Shield Advanced is available globally on various AWS services like CloudFront, Global Accelerator, and Route 53.

AWS WAF

Web Application Protection:

Focuses on protecting web applications from various web-based attacks, such as SQL injection and cross-site scripting.

Application Layer:

Operates at Layer 7 (application) of the OSI model.

Customizable Rules:

Allows users to define custom rules to filter web traffic based on various criteria, such as IP addresses, headers, and request payloads.

Integration with AWS Services:

Can be deployed in front of resources like CloudFront distributions, Application Load Balancers (ALBs), and API Gateway.

DNS and AWS Route 53

What is DNS?

DNS (Domain Name System) is a hierarchical, distributed naming system for the internet. It translates human-readable domain names (like `www.example.com`) into IP addresses (like `192.0.2.1`), which computers use to communicate. Think of it as the internet's phone book, mapping names to numerical addresses.

Route 53 is a managed DNS service provided by Amazon Web Services (AWS). It offers features like domain registration, routing policies, and health checks to ensure reliable and efficient DNS resolution.

Route 53 Routing Policies

Route 53 offers several routing policies to control how traffic is directed to your resources. Some key policies include:

Simple: Routes traffic to a single healthy resource.

Weighted : Distributes traffic between multiple resources based on their assigned weights, allowing you to prioritize certain endpoints.

Latency: Routes traffic to the region with the lowest latency for the user, optimizing performance.

Failover: Routes traffic to a secondary resource if the primary resource is unhealthy, ensuring high availability.

CloudFront (CDN)

A CDN, or Content Delivery Network, is a group of geographically distributed servers that deliver content, like websites, images, and videos, to users faster and more reliably. By caching content in servers closer to users, CDNs reduce latency, improve website performance, and enhance the overall user experience.

Key Features of a CDN

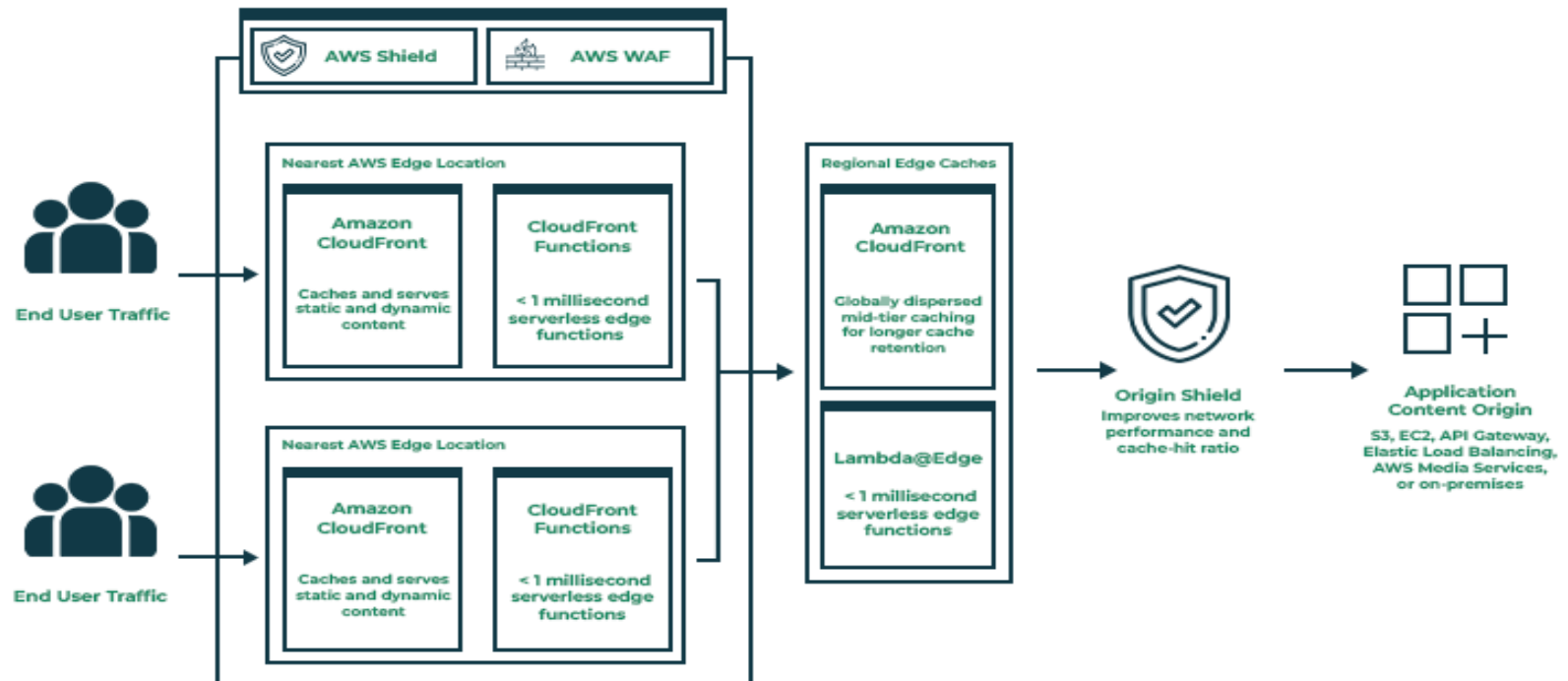
Edge Locations: CDNs use a network of edge locations (servers) strategically placed around the world.

Caching : CDNs store copies of frequently accessed content in these edge locations, allowing users to access it from a server closer to their location.

CloudFront (CDN)

HTTPS Support:

CDNs often support HTTPS, ensuring secure communication between users and servers.



How CDNs Improve Content Delivery

1. Reduced Latency:

By accessing content from a server near their location, users experience faster loading times.

2. Improved Reliability:

If one server fails, the CDN can automatically switch to another server, ensuring uninterrupted content delivery.

3. Reduced Load on Origin Servers:

CDNs offload the workload from the main server by serving content from the edge locations, preventing the main server from being overwhelmed.

4. Enhanced Security:

HTTPS support adds an extra layer of security to content delivery.

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